ALICE L WALTON SCHOOL OF MEDICINE

Benign Anorectal Disease

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Learning Objectives (3)

After completing this brick, you will be able to:

1. Using anorectal anatomy, explain the presentation of benign anorectal disease and how these presentations can overlap with malignancy.
2. Differentiate the clinical presentation and pathogenesis of hemorrhoidal disease, anal fissures, anal abscesses/fistulas, proctitis, and pruritus ani.
3. Compare an of hemorrhoidal disease, anal fissures, anal abscesses/fistulas, proctitis, and pruritus ani.

CASE CONNECTION

A 55-year-old man presents to the office with painless bleeding with bowel movements. He has to strain to pass stool. He has not had any screening procedures for colorectal cancer. His medical history and physical exam are unremarkable.

How would you proceed?

GO TO CONCLUSION ↓

Anal Canal Anatomy, Symptoms, and Overlap with Malignancy (LO 1)

Anatomy of the Anal Canal

The anal canal is approximately 4 cm long, extending from the anal verge to the anorectal junction — where the rectum passes through the levator ani (pelvic floor muscles). The **dentate (pectinate) line** is the key anatomic landmark, located roughly halfway up the canal (Figure 1). It divides the anal canal into two functionally distinct zones:

- **Above the dentate line:** lined by columnar mucosa (endoderm-derived). This tissue has **visceral innervation** — it senses stretch and pressure but not sharp pain. T rrhoids makes them more susceptible to friability and bleeding.
- **Below the dentate line:** lined by squamous epithelium (ectoderm-derived). This tissue has **somatic innervation** (pudendal nerve, S2–S4) — it is exquisitely sensitive to pain, temperature, and touch. The skin covering of external hemorrhoids provides protection from trauma and makes bleeding less likely.

This distinction explains why internal hemorrhoids bleed painlessly while thrombosed external hemorrhoids are acutely painful.

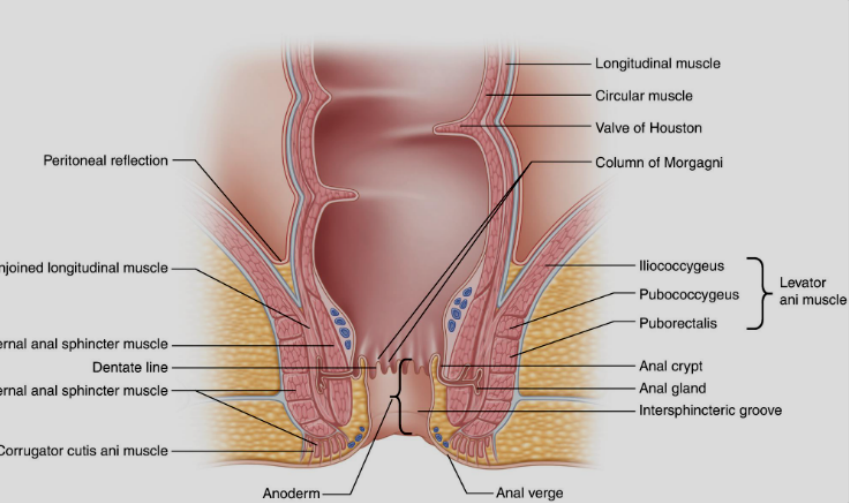


Figure 1. Anal canal anatomy

The **anal glands** open into crypts at the dentate line and extend into the intersphincteric space (Figure 1). Obstruction of these glands is the origin of perianal abscesses and fistulas.

Consider the possibility of Malignancy

While hemorrhoids and fissures are the most common causes of rectal bleeding, it is critical to remember that **colorectal cancer can present with hematochezia — passage of blood per rectum.** The presence of hemorrhoids does not exclude coexisting proximal pathology.

Colonoscopy should be pursued for any patient with:

- Rectal bleeding not typical of hemorrhoids (blood mixed with stool rather than on the surface)
- Bleeding that persists despite hemorrhoid treatment
- Associated symptoms: iron deficiency anemia, change in bowel habits, unexplained weight loss, abdominal pain
- Age-appropriate need for colorectal cancer screening
- Family history of colorectal cancer

CLINICAL REASONING

Attributing rectal bleeding to hemorrhoids without appropriate evaluation is a common reason for missed or delayed diagnosis of colorectal cancer.

Anal cancer must al e **squamous cell carcinomas** driven by persistent HPV infection (HPV 16 and 18 account for ~90% of cases). Thrombosed hemorrhoids resolve over the course of a couple weeks. A persistent, firm perianal or anal canal mass is not a "hemorrhoid", even though the patient and referring physician may describe it as such (Figure 2).

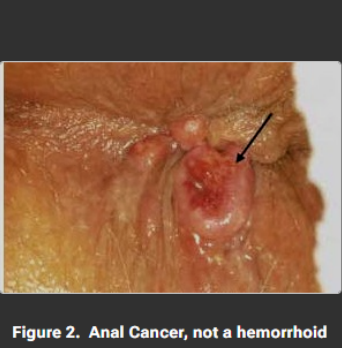


Figure 2. Anal Cancer, not a hemorrhoid

CLINICAL REASONING

Any external anal lesion that is hard, fixed, ulcerated, or fails to resolve should be biopsied to rule out malignancy, or referred to a surgeon for diagnosis.

Hemorrhoids (LO 2, 3)

Hemorrhoids are normal vascular cushions of the anal canal that contribute to continence. Hemorrh e engorged, prolapsed, or thrombosed, most often due to increased venous pressure from chronic straining or low-fiber diets.

Internal hemorrhoids arise above the dentate line. Because they are covered by viscerally innervated columnar mucosa, symptoms are typically **painless episodes of bright red bleeding**. The blood is usually separate from the stool, such as dripping into the toilet bowl or appearing on toilet tissue. As the supporting connective tissue degrades, internal hemorrhoids may **prolapse — protrude through the anal canal**, especially during defecation or straining (Figure 3). Recognizing hemorrhoid symptoms and progression is more important than memorizing a numbered system.

Increasing severity of internal hemorrhoid symptoms:

- Grade I: bleed but do not prolapse
- Grade II: prolapse with straining, reduce spontaneously
- Grade III: prolapse, require manual reduction
- Grade IV: prolapse, irreducible

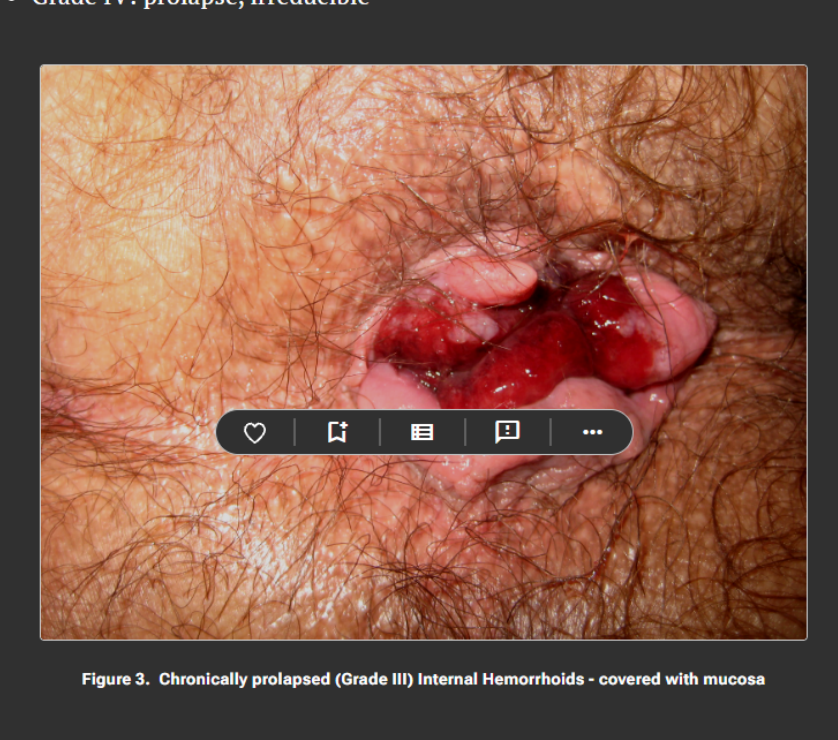


Figure 3. Chronically prolapsed (Grade III) Internal Hemorrhoids - covered with mucosa

External hemorrhoids arise below the dentate line and are covered by somatic-innervated squamous epithelium. They are typically asymptomatic unless they develop acute **thrombosis**, which causes **sudden, severe perianal pain and a firm, tender, bluish lump at the anal verge** (Figure 4). External hemorrhoids do not typically bleed because they are covered in skin which resists trauma. The pressure and swelling of thrombosed hemorrhoids may, however, cause necrosis of the overlying skin, leading to spontaneous decompression of the clot and self-limited bleeding.

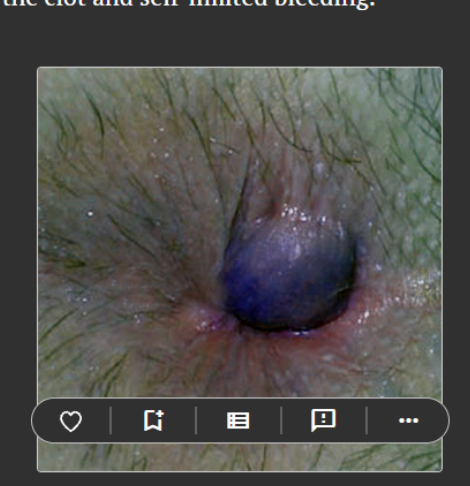


Figure 4. Thrombosed External Hemorrhoid, covered with skin

Differentiate internal and external hemorrhoids by anatomy and symptoms.

Hemorrhoid Treatment: First-line therapy for all hemorrhoids includes increased dietary fiber and adequate fluid intake. Patients should limit time on the commode to 5 minutes and avoid straining.

CLINICAL REASONING

Topical ointments and suppositories do not shrink hemorrhoidal tissue. They may provide minimal temporary relief of external discomfort, and patients may use over-the-counter pr n topical medications have not been shown to be more effective.

Symptomatic internal hemorrhoids: When bleeding or prolapsing hemorrhoids do not respond to conservative measures, **rubber band ligation** is the most effective office-based procedure. Because internal hemorrhoids are minimally sensate, no anesthetic is used. A tight rubber band is placed around the hemorrhoidal tissue **above the dentate line**, causing the tissue to necrose, slough, and heal with fibrosis that anchors the mucosa (Figure 5). **Hemorrhoidectomy** is reserved for grade III–IV disease or mixed hemorrhoidal disease refractory to less invasive treatment. Removal of both internal and external hemorrhoid tissue is an operative procedure under anesthetic (Figure 6).

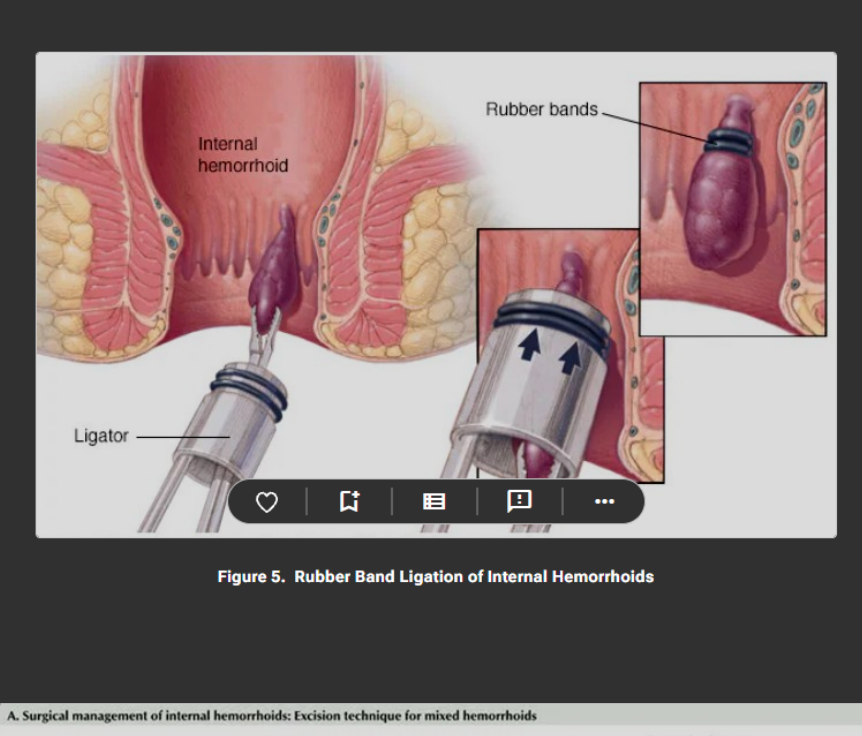


Figure 5. Rubber Band Ligation of Internal Hemorrhoids

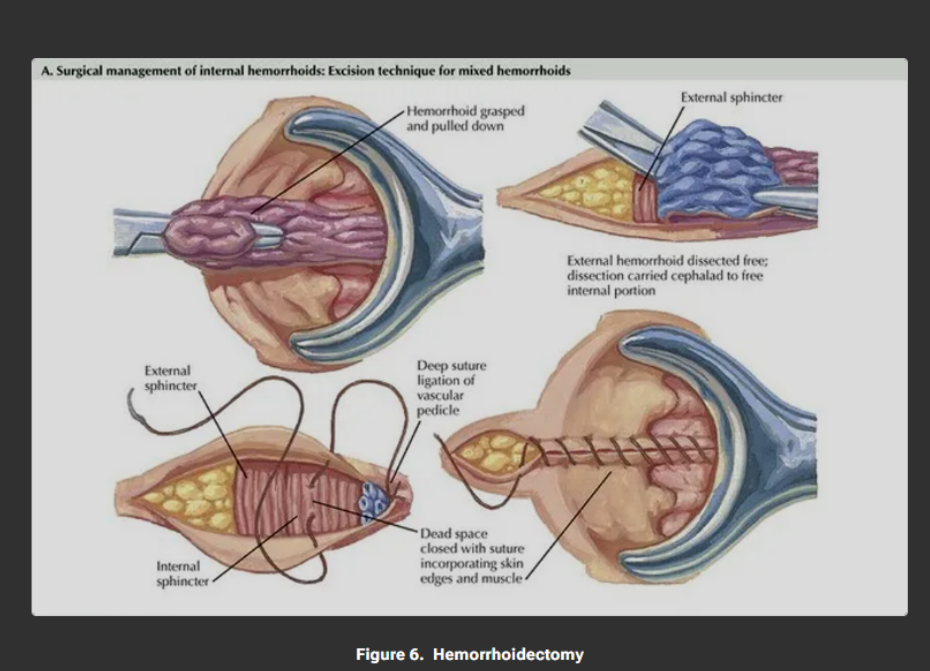

A. Surgical management of internal hemorrhoids: Excision technique for mixed hemorrhoids
Hemorrhoid grasped and pulled down
External sphincter
External hemorrhoid dissected free; dissection carried cephalad to free internal portion
External sphincter
Internal sphincter
Deep suture ligation of vascular pedicle
Dead space closed with suture incorporating skin edges and muscle

Figure 6. Hemorrhoidectomy

Thrombosed external hemorrhoids may be **excised within 3–4 days** of symptom onset if pain is severe, providing faster relief. After this period, conservative management is preferred as symptoms typically improve and procedural pain may prolong recovery (Figure 7). Excision can be performed under local anestheti ent (Figure 8). *Incision* of thrombosed hemorrhoids to remove clot is not as effective.

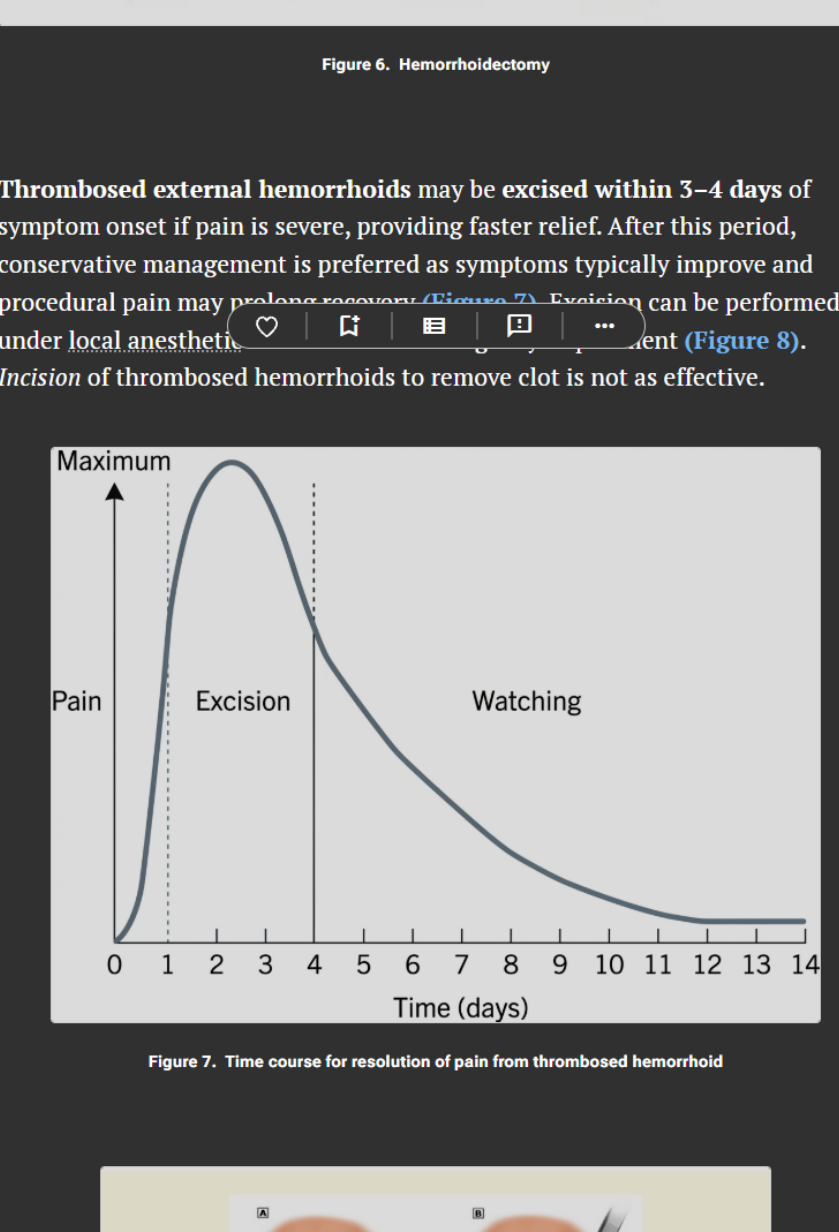


Figure 7. Time course for resolution of pain from thrombosed hemorrhoid

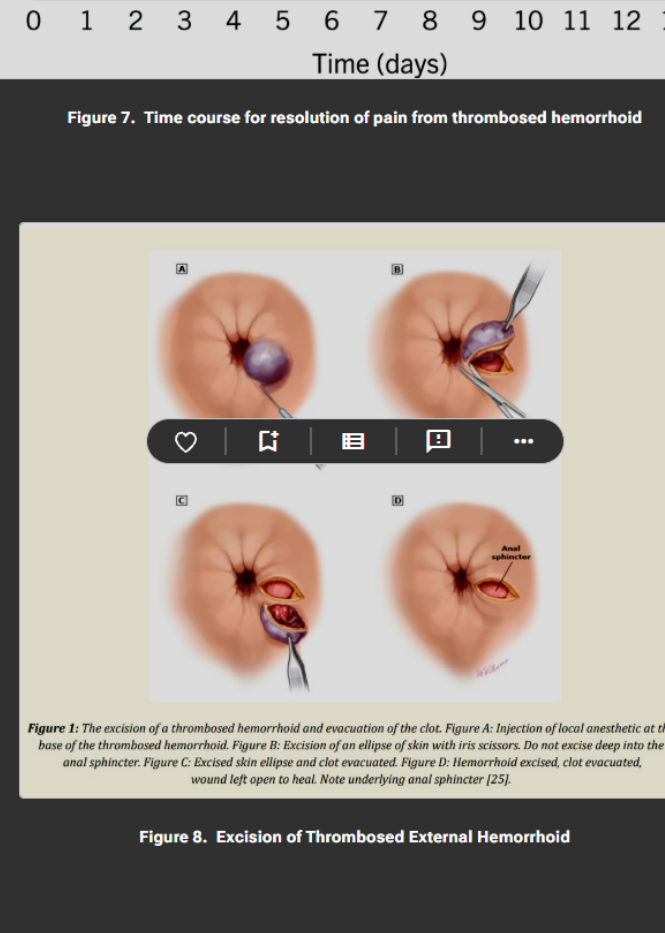


Figure 1: The excision of a thrombosed hemorrhoid and evacuation of the clot. Figure A: Injection of local anesthetic at the base of the thrombosed hemorrhoid. Figure B: Excision of an ellipse of skin with iris scissors. Do not excise deep into the anal sphincter. Figure C: Excised skin ellipse and clot evacuated. Figure D: Hemorrhoid excised, clot evacuated, wound left open to heal. Note underlying anal sphincter [25].

Figure 8. Excision of Thrombosed External Hemorrhoid

Anal Fissure (LO 2, 3)

An **anal fissure** is a **tear in the anoderm** (squamous epithelium below the dentate line), most commonly (85%) in the **posterior midline**. The initial tear is usually caused by passage of a hard stool. Because the anoderm has somatic innervation, fissures cause **sharp, tearing pain during defecation, often with a small amount of bright red blood.**

The pain triggers reflex internal sphincter spasm and hypertrophy. This spasm results in prolonged, severe pain after bowel movements and also reduces blood flow to the posterior midline (which already has the poorest blood supply in the anal canal), impairing healing. A cycle develops of pain → spasm → ischemia → non-healing. This is why acute fissures can be slow to heal and why chronic fissures develop.

Anal fissures are quit traction of the buttocks and gentle eversion of the anal mucosa will help. An acute fissure appears as a simple tear. A **chronic fissure** (>6–8 weeks) develops characteristic features: a sentinel skin tag externally, hypertrophied anal papilla internally, and visible internal sphincter fibers at the base of the ulcer (Figure 9).

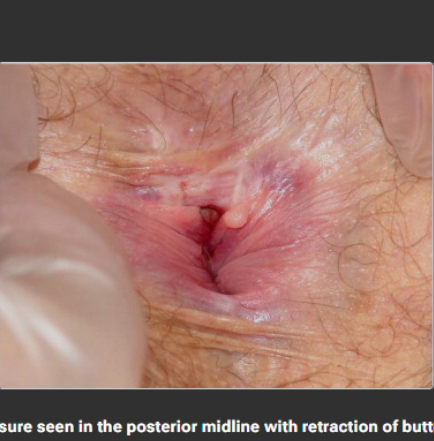
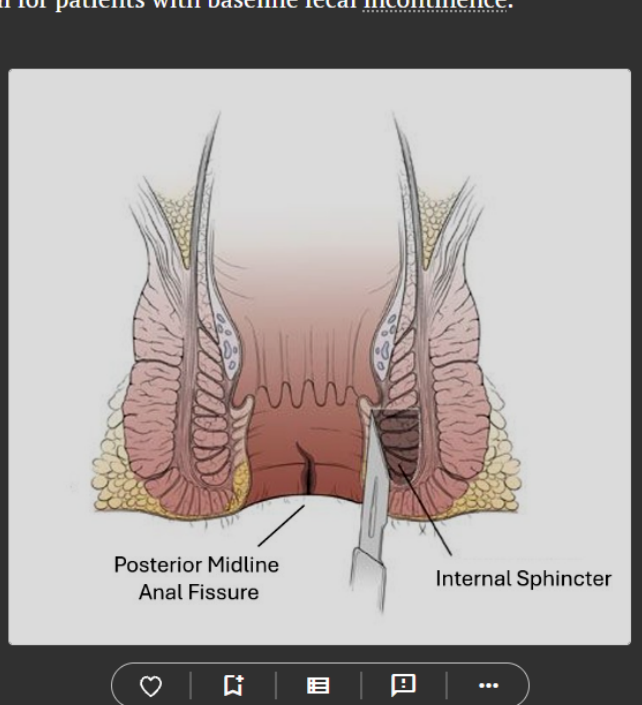


Figure 9. Chronic anal fissure seen in the posterior midline with retraction of buttocks and eversion of anal mucosa

CLINICAL REASONING

Anal fissures are usually in the posterior midline. A fissure **not in the midline** should raise suspicion for an alternate etiology such as Crohn disease, HIV, syphilis, or anal cancer.

Treatment: Acute fissures typically heal with fiber supplementation, stool softeners, and sitz baths. Both acute and chronic fissures can be treated with **topical calcium channel blockers** (e.g., nifedipine) or **topical nitroglycerin**, which relax the internal sphincter spasm and improve pain control and healing. Lateral internal sphincterotomy (LIS) is a surgical procedure that permanently reduces excessive resting internal anal sphincter tone. It is the most effective treatment for chronic anal fissures refractory to medical therapy, with excellent healing rates (Figure 10). Most patients tolerate the reduction in sphincter hypertonicity well, but the procedure may not be ideal for patients with baseline fecal incontinence.



CLINICAL REASONING

Poor healing of anal fissures results from both reduced blood flow in the posterior midline and internal sphincter spasm.

Both topical and surgical treatments are directed at the internal sphincter spasm.

Perianal Abscess and Fistula-in-Ano (LO 2, 3)

Most perianal abscesses originate from infection of the **anal glands**. The anal glands, which open at the dentate line, become obstructed, and the resulting infection spreads into the surrounding anatomic spaces (Figure 11). Abscesses are classified by location: **perianal** (most common, superficial), **ischiorectal** (ischioanal), and **intersphincteric** (Figure 12).

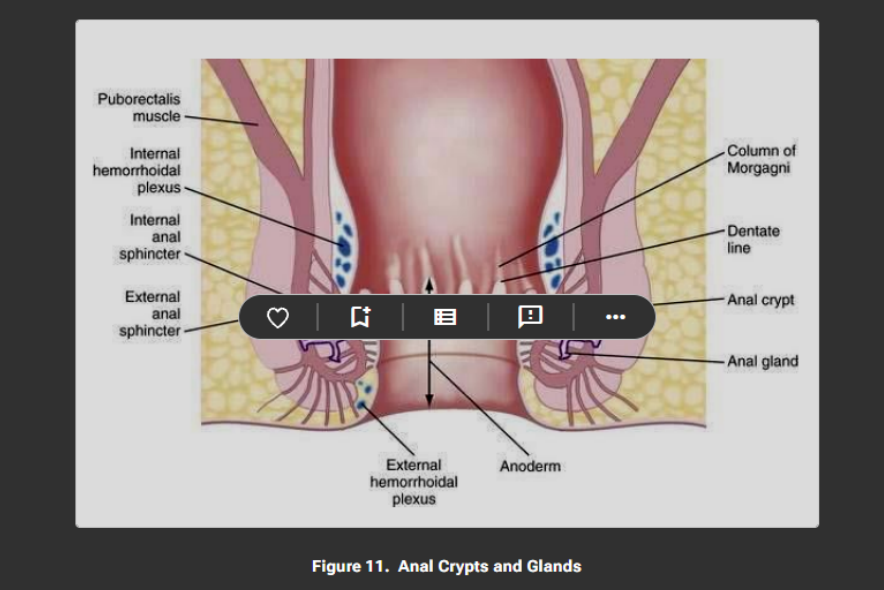


Figure 11. Anal Crypts and Glands

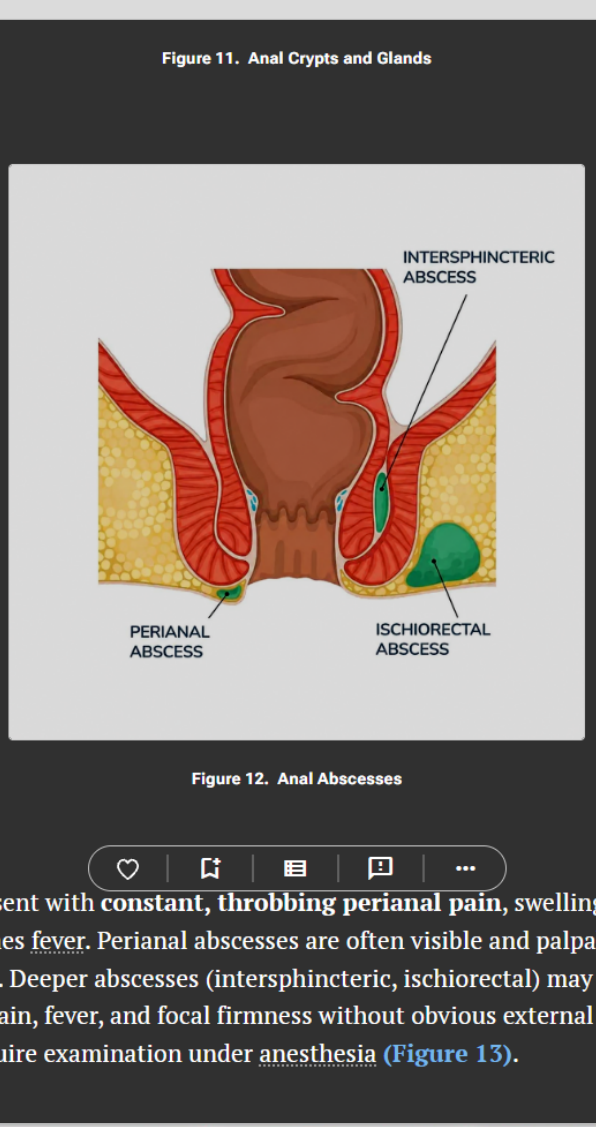


Figure 12. Anal Abscesses

Patients present with **constant, throbbing perianal pain**, swelling, erythema, and sometimes fever. Perianal abscesses are often visible and palpable on examination. Deeper abscesses (intersphincteric, ischiorectal) may present with rectal pain, fever, and focal firmness without obvious external findings and may require examination under anesthesia (Figure 13).

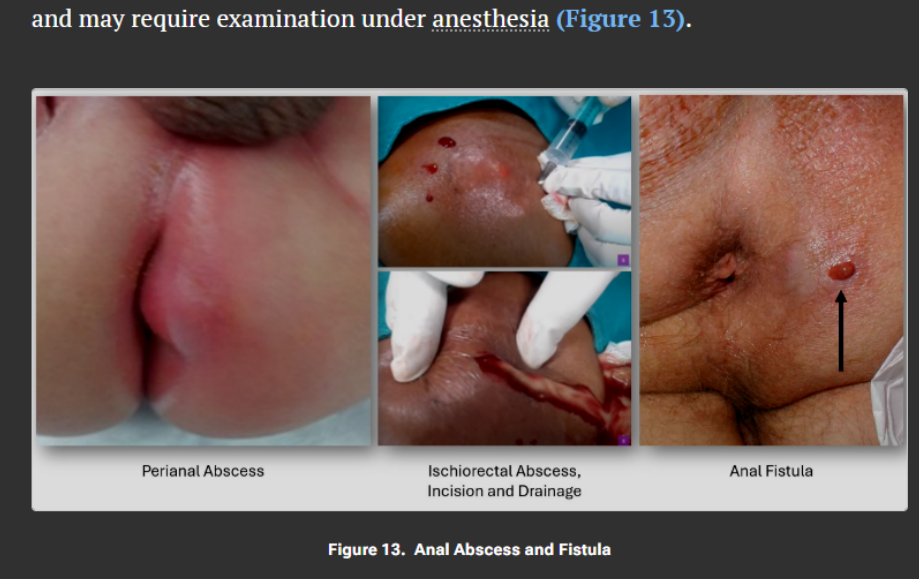


Figure 13. Anal Abscess and Fistula

CLINICAL REASONING

"Fluctuant" is the classic physical exam descriptor for an abscess, reflecting a compressible fluid collection. However, abscesses confined within deep or tight anatomic spaces may instead feel indurated or focally firm because surrounding tissues prevent obvious fluctuance.

In exam questions, l

Treatment is **prompt incision and drainage** (Figure 13, center images). Antibiotics are not needed in healthy patients — drainage alone is sufficient. Antibiotics are reserved for patients with surrounding cellulitis, systemic signs of infection, or immunosuppression.

An **anal fistula** is a tract connecting the anal canal to the perianal skin, representing a chronic form of the cryptoglandular infection. After an anal abscess drains externally, either spontaneously or by incision, failure of the internal opening to close may result in a persistent tract between the anal canal and skin (anal fistula), occurring in 30–50% of patients. Patients present with persistent or recurrent perianal drainage, often from a small opening near the anus.

Fistulas are classified by their relationship to the sphincter complex - the two most common and most simple are shown below (Figure 14). Treatment is surgical — **fistulotomy** (laying open the tract) for simple, low fistulas (Figure 15); more complex fistulas involving significant sphincter muscle require sphincter-sparing techniques to avoid fecal incontinence.

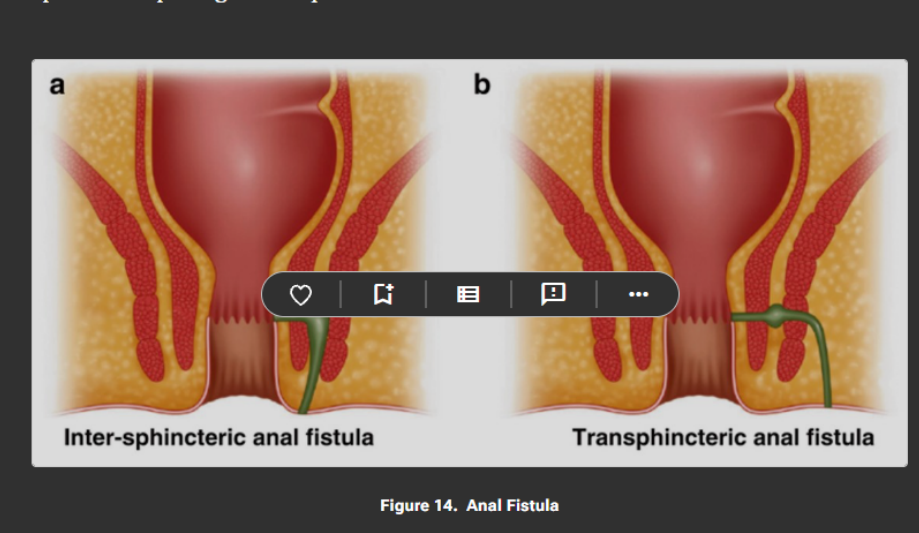


Figure 14. Anal Fistula

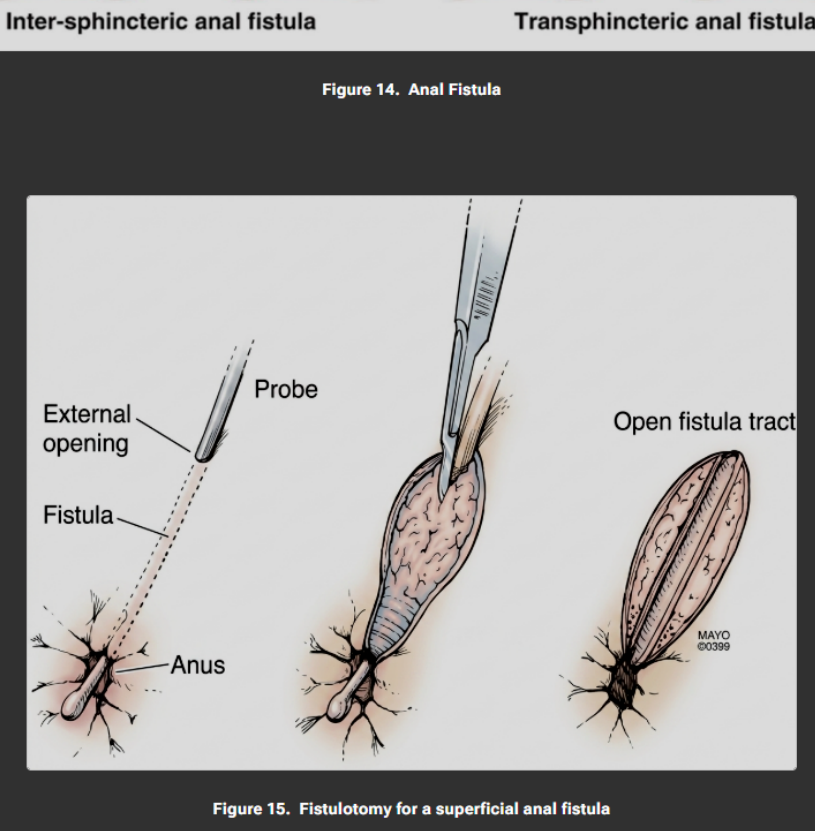


Figure 15. Fistulotomy for a superficial anal fistula

CLINICAL REASONING

Incision and drainage of a perianal abscess does not "cause" a fistula. Without treatment, the patient may become increasingly ill, and the abscess will eventually rupture through the skin with the same risk of subsequent fistula.

Proctitis (LO 2, 3)

Proctitis is inflammation of the rectal mucosa. The two major categories are **infectious** and **inflammatory**.

Infectious proctitis occurs predominantly in patients with receptive anal intercourse. The diagnosis and treatment of common pathogens (*Neisseria gonorrhoeae*, *Chlamydia trachomatis*, herpes simplex virus, and *Treponema pallidum*) is deferred to a separate unit. Patients present with anorectal pain, tenesmus, mucopurulent or bloody rectal discharge. Diagnosis involves Nucleic Acid Amplification Testing for gonorrhea and chlamydia, HSV testing, and syphilis serology.

Ulcerative proctitis is a form of ulcerative colitis limited to the rectum, accounting for up to 40% of new UC diagnoses. Patients present with rectal bleeding, urgency, and tenesmus. Diagnosis is by colonoscopy with biopsy (Figure 16). First-line treatment is **topical mesalamine** (5-ASA suppositories), which works well due to the localized nature of the inflammation.

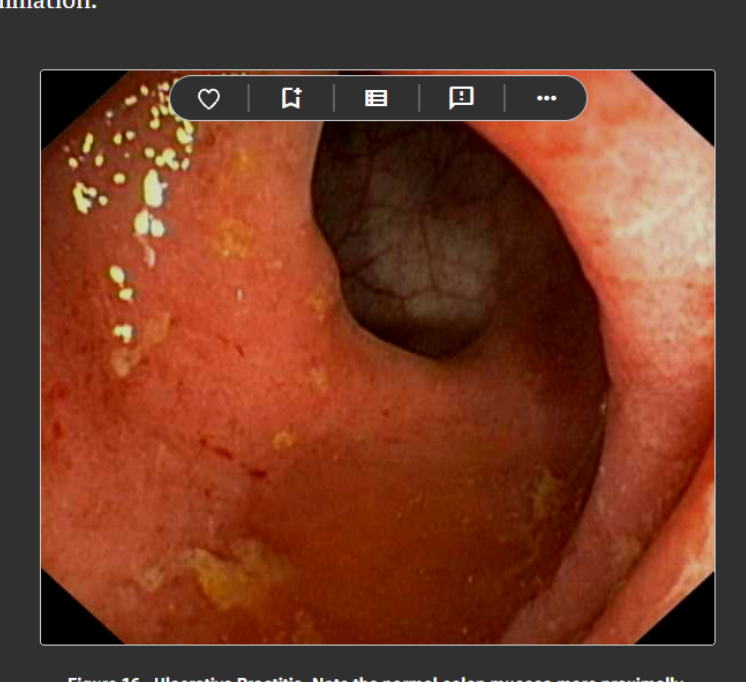


Figure 16. Ulcerative Proctitis. Note the normal colon mucosa more proximally.

CLINICAL CORRELATION

Inflammation, irritation, or distention of the rectum abnormally stimulates rectal sensory nerves and triggers spasmodic contractions of the rectal and pelvic floor muscles, creating a persistent urge to evacuate even when the rectum is empty.

Tenesmus is the resulting discomfort and sensation of needing to evacuate the rectum.

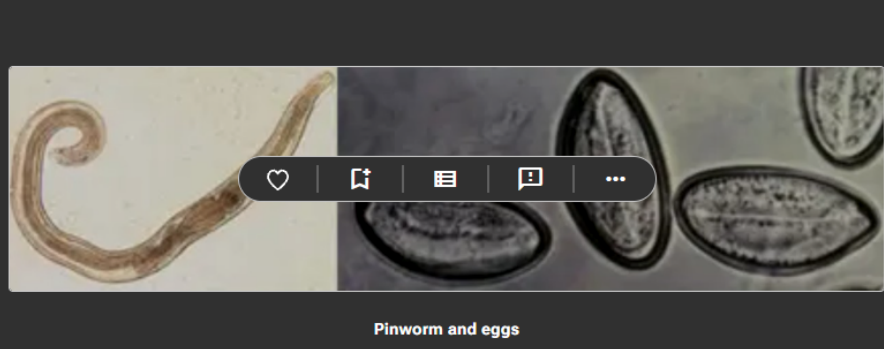
Pruritus Ani (LO 2, 3)

Pruritus ani is intense itching of the perianal skin. In many cases, associated factors are found:

- **Fecal soilage/moisture** from hemorrhoids, fissures, fistulas, or incomplete hygiene
- **Dietary irritants** — coffee, alcohol, chocolate, spicy foods, citrus
- **Dermatologic conditions** — contact dermatitis (often from wipes, soaps, or topical medications), fungal infection, psoriasis
- **Infections** — pinworms -*Enterobius vermicularis* (especially in children), candida, condylomata (HPV)
- **Neoplasia** — must be excluded, particularly in persistent or refractory cases (anal cancer)

CLINICAL REASONING

Enterobiasis (pinworm infection) is a common infection in school-age children that classically causes nocturnal perianal pruritus due to nighttime egg deposition around the anus. The diagnosis is made with the Scotch tape test performed in the early morning to collect eggs for microscopic examination.



Pinworm and eggs

Treatment. When no cause is identified, it is classified as **idiopathic pruritus ani.** Treatment focuses on **breaking the itch-scratch cycle:** gentle cleansing with water only (avoid soaps and wipes), keeping the area dry, barrier ointment (zinc oxide or petroleum jelly), and avoiding dietary triggers. A short course of low-potency topical corticosteroid (1% hydrocortisone, ≤2 weeks) can reduce acute inflammation. The disorder is notoriously refractory to treatment.

CLINICAL CORRELATION

For your interest only: **Capsaicin**, the pungent compound found in chili peppers, has shown efficacy as a topical treatment for pruritus ani — a condition that has historically been difficult to treat. Capsaicin-sensitive receptors have been identified on nerve fibers involved in itch transmission. Repeated stimulation of these receptors by topical capsaicin desensitizes the nerve endings, making them less responsive to pruritic stimuli such as histamine.

CASE CONNECTION

BACK TO INTRODUCTION ↑

You recommend a d intake, aiming to avoid straining during stent with bleeding from internal hemorrhoids, you explain there is considerable overlap between symptoms of benign and malignant disease. Since he has not had any prior colon cancer screening you proceed with colonoscopy scheduling.

At the colonoscopy appointment a month later, he states the fiber has been helpful for passage of stools. The bleeding is somewhat improved but still occurs with most BMs. His colonoscopy is negative. Prominent internal hemorrhoids are noted, so you perform hemorrhoid banding ensuring the bands are placed above the dentate line. He tolerates the procedure well and is offered follow up in 6 weeks. He instead calls your office to say his bleeding has stopped, he is passing BMs easily, and he will follow up in the future if symptoms return.

Summary

- The dentate line divides the anal canal into two functionally distinct zones: above it, columnar mucosa with visceral innervation and portal venous drainage; below it, squamous epithelium with somatic innervation (pudendal nerve) and systemic venous drainage.
- Rectal bleeding should never be reflexively attributed to hemorrhoids without appropriate evaluation; colonoscopy is indicated when bleeding is atypical, persistent, or accompanied by anemia, weight loss, change in bowel habits, or when age-appropriate cancer screening is due.
- Persistent perianal masses must be biopsied to exclude anal cancer.
- Internal hemorrhoids arise above the dentate line and present with painless bleeding or prolapse; external hemorrhoids arise below it and may thrombose, causing acute severe pain.
- First-line treatm and bowel habit modification; rubber band ligation is the most effective office procedure for symptomatic internal hemorrhoids, and excision of thrombosed external hemorrhoids is most beneficial within 3–4 days of onset.
- Anal fissures are tears in the somatic-innervated anoderm, most commonly in the posterior midline, causing sharp pain and reflex internal sphincter spasm that impairs healing. Treatment targets sphincter hypertonicity — topical calcium channel blockers or nitroglycerin for chronic fissures, and lateral internal sphincterotomy for refractory cases.
- Most perianal abscesses arise from obstruction of anal glands at the dentate line and require prompt incision and drainage; many will result in anal fistula, a chronic tract connecting the anal canal to the perianal skin that requires surgical treatment.
- Proctitis presents with rectal pain, urgency, tenesmus, and bleeding, and is classified as infectious (commonly STIs) or inflammatory (ulcerative proctitis, treated with topical mesalamine).
- Pruritus ani is commonly associated with fecal soilage, dietary irritants or dermatologic conditions, and is managed by gentle hygiene, moisture control, and barrier protection, with malignancy excluded in refractory cases.

Review Questions

1. A 32-year-old male presents to your office with concerns about something bulging from his anal canal. Blood will drip into the toilet during BMs. He has no pain. Physical exam reveals a small mass of tissue, covered with mucosa, prolapsing out the anal canal. You can reduce it with digital rectal exam, without any significant pain. What is the most likely etiology?

- A. ○ anal fissure
- B. ○ anal fistula
- C. ○ prolapsing internal hemorrhoid
- D. ○ prolapsing external hemorrhoid
- E. ○ thrombosed internal hemorrhoid
- F. ○ thrombosed external hemorrhoid

Explanation (requires correct answer)

2. Which of the following is the correct description of an external hemorrhoid?

- A. ○ above dentate line, covered with mucosa, visceral sensation
- B. ○ above dentate line, covered with skin, somatic sensation
- C. ○ below dentate line, covered with mucosa, visceral sensation
- D. ○ below dentate line, covered with skin, somatic sensation

Explanation (requires correct answer)

3. A 22-year-old female has had 5 days of anal pain with a swollen, tender perianal mass. On examination it is at the edge of the anal canal, covered in erythematous skin, and is exquisitely tender. What is your treatment?

- A. ○ antibiotics alone
- B. ○ banding
- C. ○ excision
- D. ○ incision

Explanation (requires correct answer)

How would you rate this Brick?

👍 👎

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☐ Using anorectal anatomy, explain the presentation of benign anorectal disease and how these presentations can overlap with malign

Explanation (requires correct answer)

How would you rate this Brick?

👍 👎

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

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02 ☐ Differentiate the clinical presentation and pathogenesis of hemorrhoidal disease, anal fissures, anal abscesses/fistulas, proctitis, and pruritus ani.

03 ☐ Compare and contrast the diagnosis and treatment of hemorrhoidal disease, anal fissures, anal abscesses/fistulas, proctitis, and pruritus ani.

Objective Confidence: 0/3